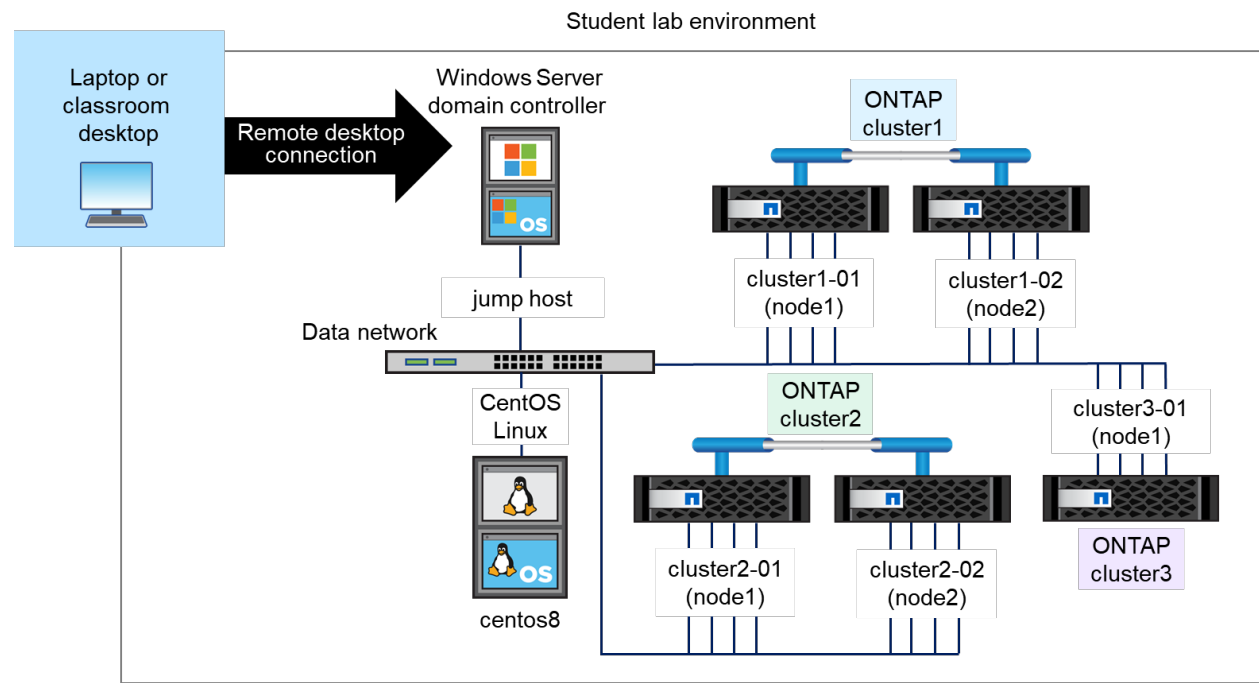


Module 1: SnapMirror Fundamentals

Exercise Equipment Diagram



System	Host Name	IP Addresses	User Name	Password
Windows Server 2022 Datacenter	JUMPHOST	192.168.0.5	DEMO\Administrator	Netapp1!
ONTAP cluster-management LIF (Cluster1)	Cluster1	192.168.0.101	admin (case-sensitive)	Netapp1!
node 1 (Cluster1)	cluster1-01-console	192.168.0.111	admin (case-sensitive)	Netapp1!
node 2 (Cluster1)	cluster1-02-console	192.168.0.112	admin (case-sensitive)	Netapp1!
ONTAP cluster-management LIF (Cluster2)	Cluster2	192.168.0.102	admin (case-sensitive)	Netapp1!
node 1 (Cluster2)	cluster2-01-console	192.168.0.113	admin (case-sensitive)	Netapp1!
node 2 (Cluster2)	cluster2-02-console	192.168.0.114	admin (case-sensitive)	Netapp1!
ONTAP cluster-management LIF (Cluster3)	Cluster3	192.168.0.103	admin (case-sensitive)	Netapp1!
node 1 (Cluster3)	cluster3-01-console	192.168.0.115	admin (case-sensitive)	Netapp1!
Linux Server	centos8	192.168.0.21	root	Netapp1!

Exercise 1: Configuring Cluster Peering and Storage VM Peering

In this exercise, you use NetApp ONTAP System Manager to set up the prerequisite configurations for data protection between Cluster1 (the primary ONTAP cluster) and Cluster2 (the secondary ONTAP cluster).

Objectives

This exercise focuses on enabling you to do the following:

- Create intercluster LIFs for both clusters
- Configure cluster peering
- Configure storage VM (storage virtual machine, also known as SVM) peering
- Review the exercise environment

Task 1: Create Intercluster LIFs on Cluster1

In this task, you implement the network connectivity to enable peering relationships between clusters. In the following steps, you create intercluster LIFs for Cluster1 and Cluster2.

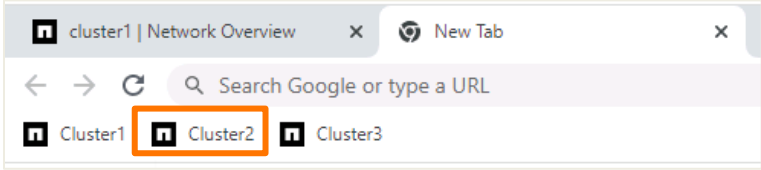
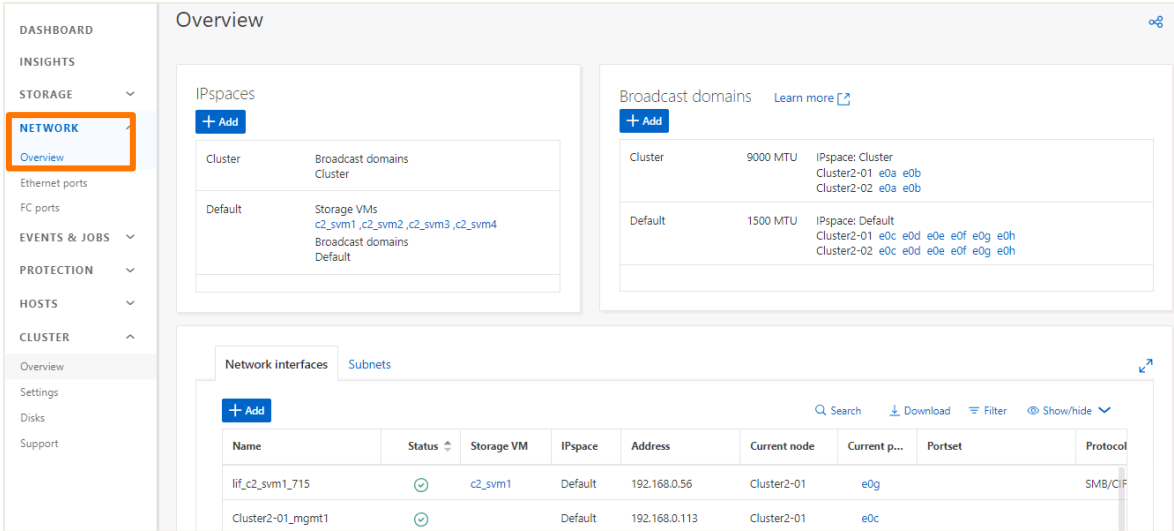
Step	Action																																																	
1-1	 When you use the ONTAP CLI to perform the exercises, you often switch between PuTTY sessions for each cluster. To help you avoid mistakes, exercise steps that require you to switch between clusters include the corresponding cluster label. Steps that occur on Cluster1 begin with a Cluster1 label. Steps that occur on Cluster2 begin with a Cluster2 label. Steps that occur on Cluster3 begin with a Cluster3 label.																																																	
1-2	In the System Manager navigation pane for Cluster1, select Network > Overview. DASHBOARD INSIGHTS STORAGE NETWORK Overview Ethernet ports FC ports EVENTS & JOBS PROTECTION HOSTS CLUSTER Overview Settings Disks Support IPspaces + Add <table><tr><td>Cluster</td><td>Broadcast domains</td></tr><tr><td>Cluster</td><td>Cluster</td></tr><tr><td>Default</td><td>Storage VMs</td></tr><tr><td>c3_svm2, c3_svm3</td><td>Broadcast domains</td></tr><tr><td>Default, Default-1</td><td></td></tr></table> Broadcast domains + Add Learn more <table><tr><td>Cluster</td><td>9000 MTU</td><td>IPspace: Cluster</td></tr><tr><td>Default</td><td>1500 MTU</td><td>IPspace: Default</td></tr><tr><td>Default-1</td><td>1500 MTU</td><td>IPspace: Default</td></tr><tr><td></td><td></td><td>Cluster3-01 e0c e0d e0e e0f e0g e0h</td></tr><tr><td></td><td></td><td>Cluster3-01 e0a e0b</td></tr></table> Network interfaces Subnets + Add Q Search Download Filter Show/hide <table><tr><th>Name</th><th>Status</th><th>Storage VM</th><th>IPspace</th><th>Address</th><th>Current node</th><th>Current p...</th><th>Portset</th></tr><tr><td>Cluster3-01_mgmt1</td><td>✓</td><td></td><td>Default</td><td>192.168.0.115</td><td>Cluster3-01</td><td>e0c</td><td></td></tr><tr><td>lif_c3_svm2_222</td><td>✓</td><td>c3_svm2</td><td>Default</td><td>192.168.0.53</td><td>Cluster3-01</td><td>e0h</td><td></td></tr></table>	Cluster	Broadcast domains	Cluster	Cluster	Default	Storage VMs	c3_svm2, c3_svm3	Broadcast domains	Default, Default-1		Cluster	9000 MTU	IPspace: Cluster	Default	1500 MTU	IPspace: Default	Default-1	1500 MTU	IPspace: Default			Cluster3-01 e0c e0d e0e e0f e0g e0h			Cluster3-01 e0a e0b	Name	Status	Storage VM	IPspace	Address	Current node	Current p...	Portset	Cluster3-01_mgmt1	✓		Default	192.168.0.115	Cluster3-01	e0c		lif_c3_svm2_222	✓	c3_svm2	Default	192.168.0.53	Cluster3-01	e0h	
Cluster	Broadcast domains																																																	
Cluster	Cluster																																																	
Default	Storage VMs																																																	
c3_svm2, c3_svm3	Broadcast domains																																																	
Default, Default-1																																																		
Cluster	9000 MTU	IPspace: Cluster																																																
Default	1500 MTU	IPspace: Default																																																
Default-1	1500 MTU	IPspace: Default																																																
		Cluster3-01 e0c e0d e0e e0f e0g e0h																																																
		Cluster3-01 e0a e0b																																																
Name	Status	Storage VM	IPspace	Address	Current node	Current p...	Portset																																											
Cluster3-01_mgmt1	✓		Default	192.168.0.115	Cluster3-01	e0c																																												
lif_c3_svm2_222	✓	c3_svm2	Default	192.168.0.53	Cluster3-01	e0h																																												

Step	Action																														
1-3	In the Network interfaces tab, click Add. Network interfacesSubnets + Add <table><thead><tr><th>Name</th><th>Status</th><th>Storage VM</th><th>IPspace</th><th>Address</th><th>Current node</th></tr></thead><tbody><tr><td>c1_svm2_nas_lif1</td><td>✓</td><td>c1_svm2</td><td>Default</td><td>192.168.0.61</td><td>Cluster1-02</td></tr><tr><td>cluster_mgmt</td><td>✓</td><td></td><td>Default</td><td>192.168.0.101</td><td>Cluster1-01</td></tr><tr><td>c1_svm3_nas_lif1</td><td>✓</td><td>c1_svm3</td><td>Default</td><td>192.168.0.62</td><td>Cluster1-01</td></tr><tr><td>Cluster1-02_clus1</td><td>✓</td><td></td><td>Cluster</td><td>169.254.24.237</td><td>Cluster1-02</td></tr></tbody></table>	Name	Status	Storage VM	IPspace	Address	Current node	c1_svm2_nas_lif1	✓	c1_svm2	Default	192.168.0.61	Cluster1-02	cluster_mgmt	✓		Default	192.168.0.101	Cluster1-01	c1_svm3_nas_lif1	✓	c1_svm3	Default	192.168.0.62	Cluster1-01	Cluster1-02_clus1	✓		Cluster	169.254.24.237	Cluster1-02
Name	Status	Storage VM	IPspace	Address	Current node																										
c1_svm2_nas_lif1	✓	c1_svm2	Default	192.168.0.61	Cluster1-02																										
cluster_mgmt	✓		Default	192.168.0.101	Cluster1-01																										
c1_svm3_nas_lif1	✓	c1_svm3	Default	192.168.0.62	Cluster1-01																										
Cluster1-02_clus1	✓		Cluster	169.254.24.237	Cluster1-02																										
1-4	In the Add network interface dialog box, specify the following values: - Interface Role: Intercluster - Name: Intercluster_LIF1 - Home Node: Cluster1-01 - IP Address: 192.168.0.105 - Subnet Mask: 24 (automatically generated) Add network interface INTERFACE ROLE ⓘ ☐ Data ☒ Intercluster ☐ Storage VM management NAME Intercluster_LIF1 HOME NODE Cluster1-01 SUBNET Without a subnet IP ADDRESS 192.168.0.105 SUBNET MASK 24 GATEWAY Add optional gateway HOME PORT Automatically select a home port (recommended) SaveCancel																														
1-5	Click Save.																														

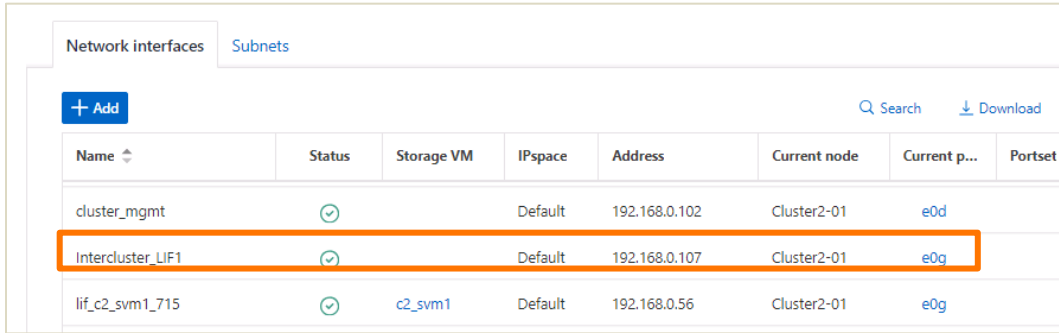
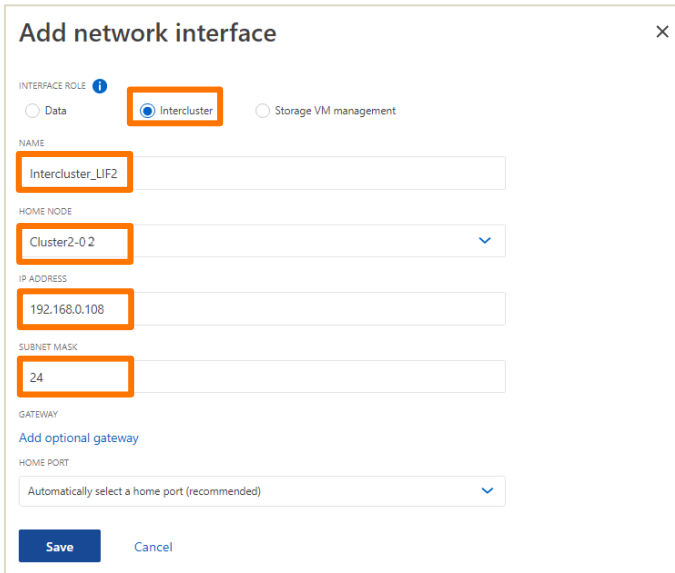
Step	Action																																				
1-6	 ONTAP software can automatically select a port for new network interfaces. Therefore, the port selection option is set to Automatically select a home port (recommended). To manually choose a port, you can click and select it from the Home Port drop-down menu. You can move or rehome a port through the UI.																																				
1-7	Verify that Intercluster_LIF1 appears in the Network interfaces list. Network interfacesSubnets + Add <table><thead><tr><th>Name</th><th>Status</th><th>Storage VM</th><th>IPspace</th><th>Address</th><th>Current node</th></tr></thead><tbody><tr><td>Cluster1-02_clus2</td><td></td><td></td><td>Cluster</td><td>169.254.96.93</td><td>Cluster1-02</td></tr><tr><td>Cluster1-02_mgmt1</td><td></td><td></td><td>Default</td><td>192.168.0.112</td><td>Cluster1-02</td></tr><tr><td>c1_svm1_nas_lif1</td><td></td><td>c1_svm1</td><td>Default</td><td>192.168.0.60</td><td>Cluster1-01</td></tr><tr><td>Intercluster_LIF1</td><td></td><td></td><td>Default</td><td>192.168.0.105</td><td>Cluster1-01</td></tr><tr><td>Cluster1-01_mgmt1</td><td></td><td></td><td>Default</td><td>192.168.0.111</td><td>Cluster1-01</td></tr></tbody></table>	Name	Status	Storage VM	IPspace	Address	Current node	Cluster1-02_clus2			Cluster	169.254.96.93	Cluster1-02	Cluster1-02_mgmt1			Default	192.168.0.112	Cluster1-02	c1_svm1_nas_lif1		c1_svm1	Default	192.168.0.60	Cluster1-01	Intercluster_LIF1			Default	192.168.0.105	Cluster1-01	Cluster1-01_mgmt1			Default	192.168.0.111	Cluster1-01
Name	Status	Storage VM	IPspace	Address	Current node																																
Cluster1-02_clus2			Cluster	169.254.96.93	Cluster1-02																																
Cluster1-02_mgmt1			Default	192.168.0.112	Cluster1-02																																
c1_svm1_nas_lif1		c1_svm1	Default	192.168.0.60	Cluster1-01																																
Intercluster_LIF1			Default	192.168.0.105	Cluster1-01																																
Cluster1-01_mgmt1			Default	192.168.0.111	Cluster1-01																																
1-8	 For fault tolerance, each node of the cluster must have an intercluster LIF. In the next steps, you create an intercluster LIF named Intercluster_LIF2 on node 2.																																				
1-9	From the Network interfaces tab, click Add .																																				

Task 2: Prepare the Storage Environment on Cluster2 as the Secondary Target

In this task, you repeat the steps that you used in Task 1 to create an intercluster LIF on the secondary cluster (Cluster2).

Step	Action
2-1	In your browser, open System Manager for Cluster2 in a new tab, ignore any security warning that might appear, and proceed to log in. 
2-2	Log in to System Manager for Cluster2 by using the following credentials: ▪ User name: admin ▪ Password: Netapp1!
2-3	In the System Manager navigation pane, select Network > Overview. 
2-4	 Alternatively, you can create intercluster LIFs from the Protection > Overview pane.

Step	Action																																								
2-5	In the Overview > Network interfaces tab, click Add. Network interfacesSubnets + Add Search Download <table><thead><tr><th>Name</th><th>Status</th><th>Storage VM</th><th>IPspace</th><th>Address</th><th>Current node</th><th>Current p...</th><th>Portset</th></tr></thead><tbody><tr><td>lif_c2_svm1_715</td><td></td><td>c2_svm1</td><td>Default</td><td>192.168.0.56</td><td>Cluster2-01</td><td>e0g</td><td></td></tr><tr><td>Cluster2-01_mgmt1</td><td></td><td></td><td>Default</td><td>192.168.0.113</td><td>Cluster2-01</td><td>e0c</td><td></td></tr><tr><td>Cluster2-01_clus2</td><td></td><td></td><td>Cluster</td><td>169.254.74.112</td><td>Cluster2-01</td><td>e0b</td><td></td></tr><tr><td>Cluster2-01_clus1</td><td></td><td></td><td>Cluster</td><td>169.254.187.153</td><td>Cluster2-01</td><td>e0a</td><td></td></tr></tbody></table>	Name	Status	Storage VM	IPspace	Address	Current node	Current p...	Portset	lif_c2_svm1_715		c2_svm1	Default	192.168.0.56	Cluster2-01	e0g		Cluster2-01_mgmt1			Default	192.168.0.113	Cluster2-01	e0c		Cluster2-01_clus2			Cluster	169.254.74.112	Cluster2-01	e0b		Cluster2-01_clus1			Cluster	169.254.187.153	Cluster2-01	e0a	
Name	Status	Storage VM	IPspace	Address	Current node	Current p...	Portset																																		
lif_c2_svm1_715		c2_svm1	Default	192.168.0.56	Cluster2-01	e0g																																			
Cluster2-01_mgmt1			Default	192.168.0.113	Cluster2-01	e0c																																			
Cluster2-01_clus2			Cluster	169.254.74.112	Cluster2-01	e0b																																			
Cluster2-01_clus1			Cluster	169.254.187.153	Cluster2-01	e0a																																			
2-6	In the Add network interface dialog box, specify the following values: - Interface Role: Intercluster - Name: Intercluster_LIF1 - Home Node: Cluster2-01 - IP Address: 192.168.0.107 - Subnet Mask: 24 (automatically generated) Add network interface INTERFACE ROLE ⓘ ☐ Data ☒ Intercluster ☐ Storage VM management NAME Intercluster_LIF1 HOME NODE Cluster2-01 IP ADDRESS 192.168.0.107 SUBNET MASK 24 GATEWAY Add optional gateway HOME PORT Automatically select a home port (recommended) Save Cancel																																								
2-7	Click Save .																																								

Step	Action
2-8	Verify that Intercluster_LIF1 appears in the Network interfaces list. 
2-9	In the Network interfaces tab, click Add.
2-10	In the Add network interface dialog box, specify the following values: - Interface Role: Intercluster - Name: Intercluster_LIF2 - Home Node: Cluster2-02 - IP Address: 192.168.0.108 - Subnet Mask: 24 (automatically generated) 
2-11	Click Save.

Step

Action

2-12

Verify that the two new intercluster LIFs (Intercluster_LIF1 and Intercluster_LIF2) are listed for Cluster2.

Network interfaces

Subnets

+ Add

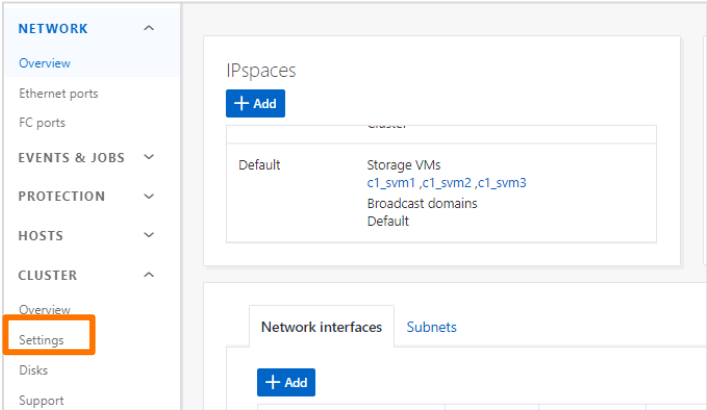
Search

Download

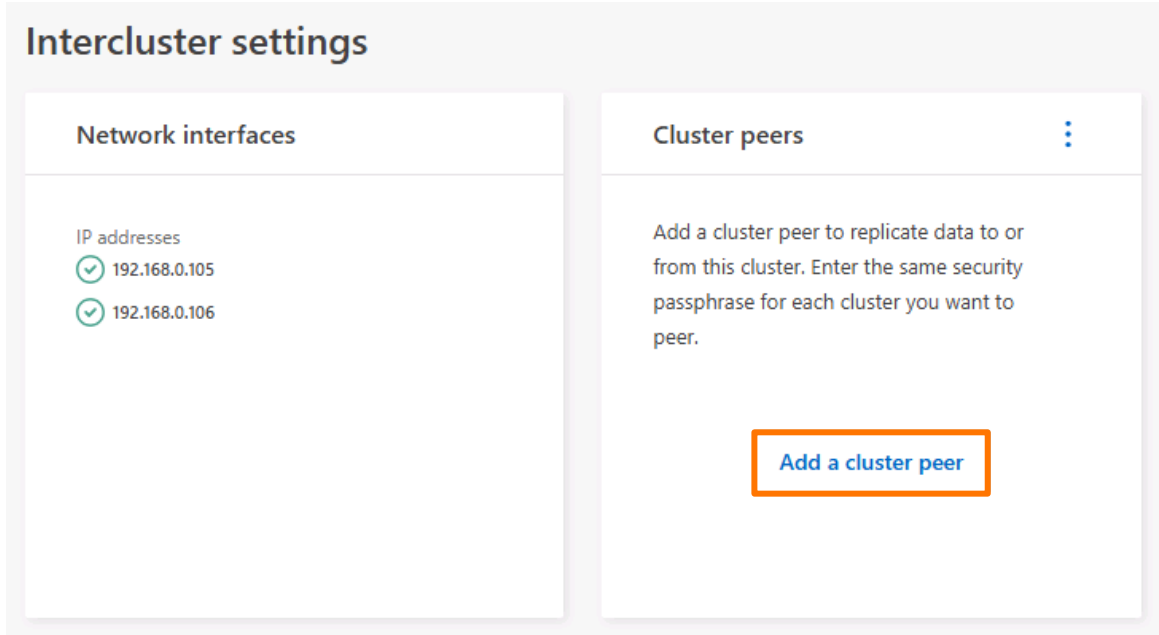
Name	Status	Storage VM	IPspace	Address	Current node	Current p...	Portset
cluster_mgmt	✓		Default	192.168.0.102	Cluster2-01	e0d	
Intercluster_LIF1	✓		Default	192.168.0.107	Cluster2-01	e0d	
Intercluster_LIF2	✓		Default	192.168.0.108	Cluster2-02	e0e	
lif_c2_svm1_127	✓	c2_svm1	Default	192.168.0.56	Cluster2-01	e0c	
lif_c2_svm3_460	✓	c2_svm3	Default	192.168.0.50	Cluster2-01	e0e	

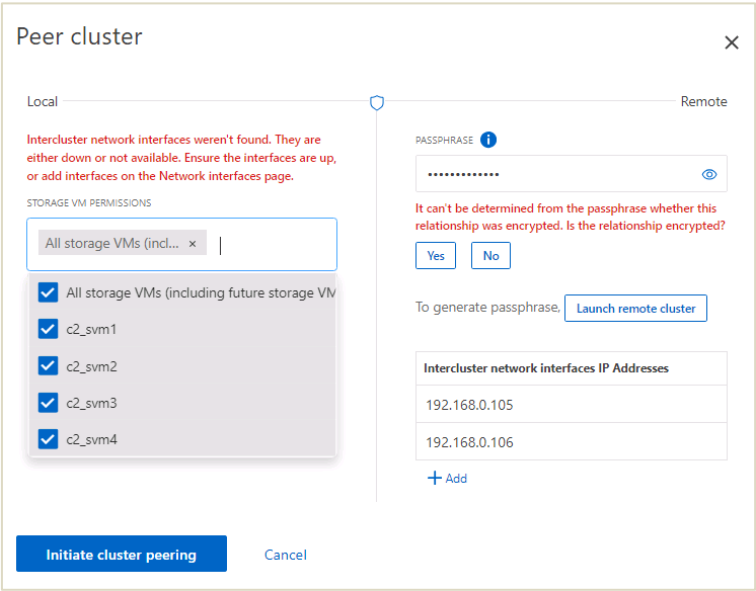
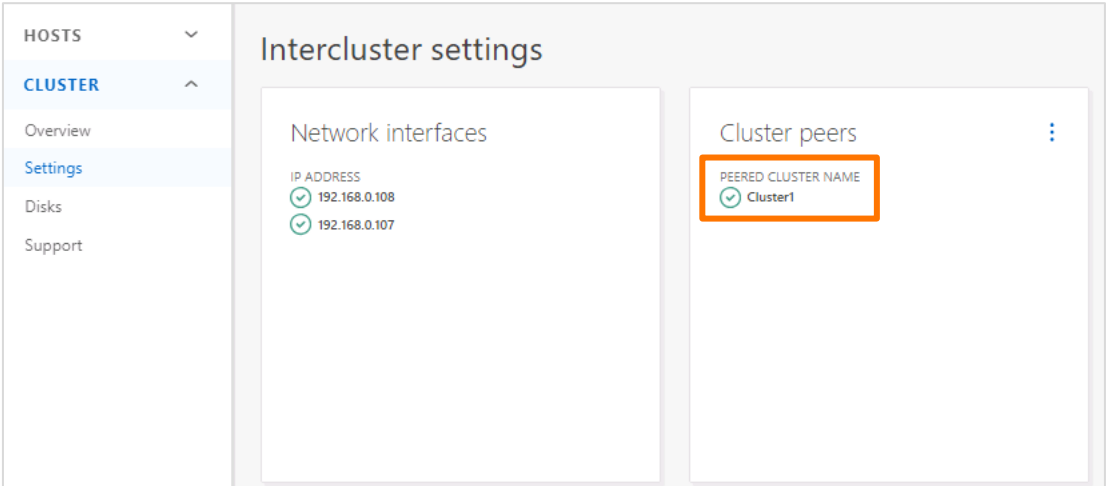
Task 3: Configure Cluster Peering

Cluster peering is a requirement of intercluster communication. In this task, you configure a peer relationship between Cluster1 and Cluster2. You must create the peer relationship on Cluster1 and then use the same steps to create the relationship on Cluster2.

Step	Action
3-1	On Cluster1, in the System Manager navigation pane, select Cluster > Settings. 
3-2	 Alternatively, you can create peer clusters from the Protection > Overview pane.
3-3	Perform the following steps: - Scroll down to the Intercluster settings section. - In the Intercluster settings > Network interfaces pane, verify the intercluster LIF details for Cluster1. - In the Intercluster settings > Cluster peers pane, click “Add a peer cluster”

Step	Action
	Intercluster settings Network interfaces IP addresses ✓ 192.168.0.105 ✓ 192.168.0.106 Cluster peers Add a cluster peer to replicate data to or from this cluster. Enter the same security passphrase for each cluster you want to peer. Add a cluster peer
3-4	In the Peer cluster dialog box, perform the following steps: - For Passphrase, enter dpadmincourse and then click outside the text box. - If you want to verify the passphrase, in the Passphrase text box, click the eyeball icon. - In response to the question “Is the relationship encrypted?”, click Yes. - Click Add. - In the Intercluster network interfaces IP Addresses field, enter 192.168.0.107 and 192.168.0.108. - Click outside the text box. - In the Storage VM Permissions field, select the All storage VMs (including future storage VMs) checkbox. Add a cluster peer Local Storage VM permissions All storage VMs (including future storage) ✓ ✓ c1_svm1 ✓ c1_svm2 ✓ c1_svm3 Remote Passphrase ? dpadmincourse 👁 To generate passphrase: Launch remote cluster Intercluster network interfaces IP Addresses 192.168.0.107 192.168.0.108 + Add Initiate cluster peering Cancel

Step	Action
3-5	 The passphrase must be at least eight characters. You can use any passphrase, but you must use the same passphrase on both clusters.
3-6	Click Initiate cluster peering .
3-7	Perform the following steps for Cluster2: - In the System Manager navigation pane, select Cluster > Settings. - In the Intercluster settings > Cluster peers pane, click Add a cluster peer. 
3-8	In the Peer cluster dialog box, perform the following tasks: - For Passphrase, enter dpadmincourse. - For relationship encryption, click Yes. - Add two Intercluster network interfaces IP Addresses: 192.168.0.105 and 192.168.0.106. - In the Storage VM Permissions field, select the All storage VMs (including future storage VMs) checkbox.

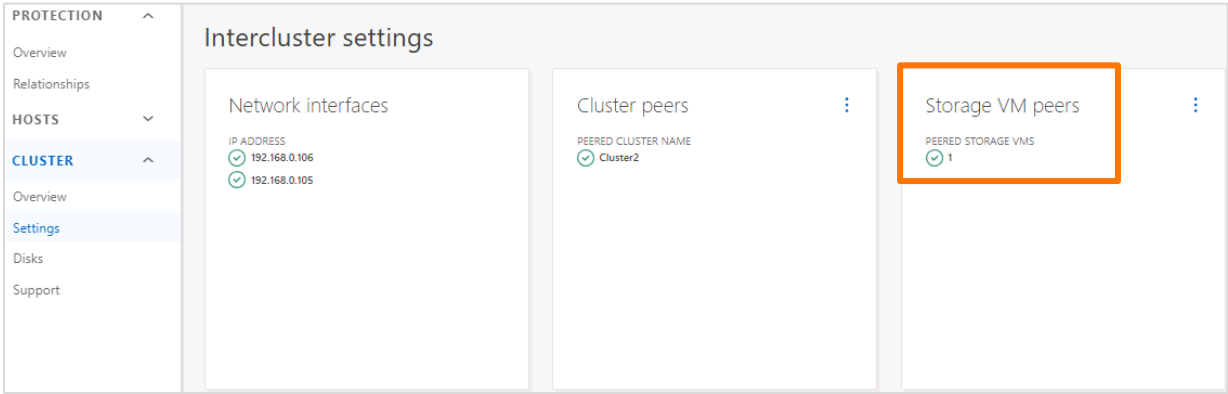
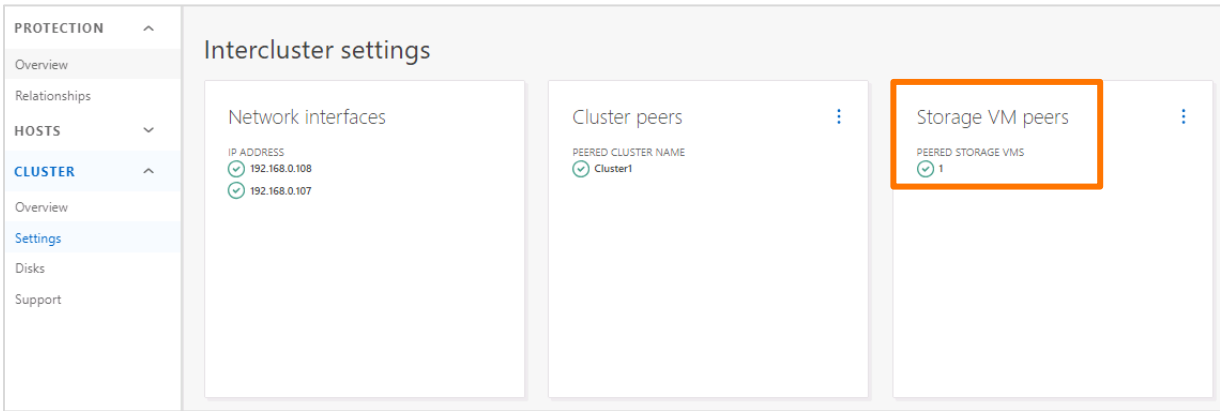
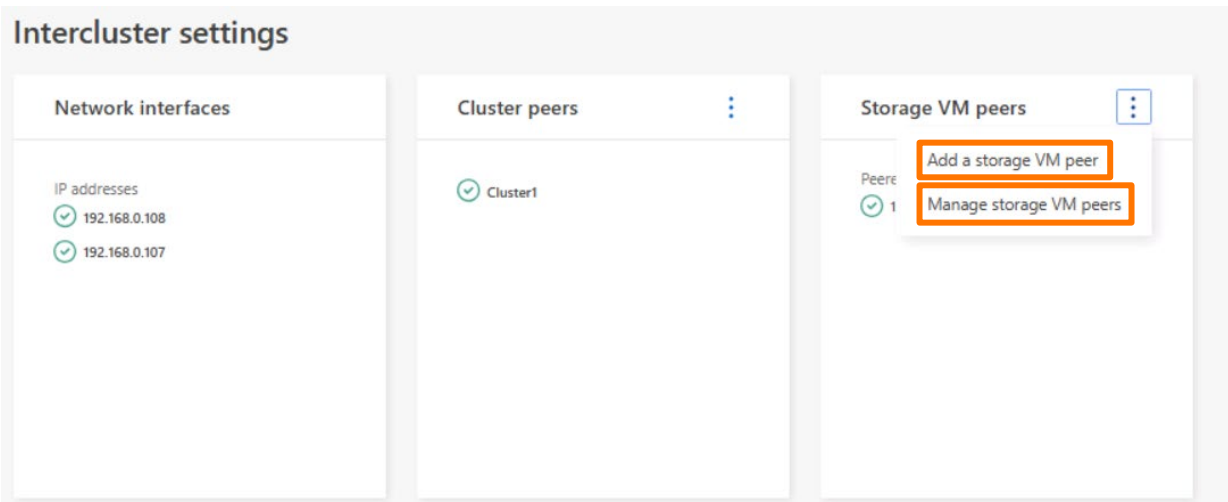
Step	Action
	
3-9	Click Initiate cluster peering and wait a few seconds for the peering operation to finish.
3-10	In the Intercluster Settings > Cluster Peers pane, verify that Cluster1 appears, indicating that the cluster peering is successful on Cluster2.
	
3-11	On Cluster1, in the System Manager navigation pane, select Cluster > Settings .
3-12	In the Intercluster settings > Cluster peers pane, wait for the status icon to turn green to confirm that cluster peering is successful on Cluster1.

Step	Action
	PROTECTION ^ Overview Relationships HOSTS v CLUSTER ^ Overview Settings Disks Support Intercluster settings Network interfaces IP ADDRESS ✓ 192.168.0.106 ✓ 192.168.0.105 Cluster peers PEERED CLUSTER NAME ✓ Cluster2
3-13	In the CLI for both Cluster1 and Cluster2, run the following commands to review the cluster peering status: • <code>cluster peer show</code> • <code>cluster peer health show</code> • <code>cluster peer ping</code>

Task 4: Configure Storage VM Peering

In this task, you create a peer relationship between c1_svm3 on Cluster1 and c2_svm1 on Cluster2.

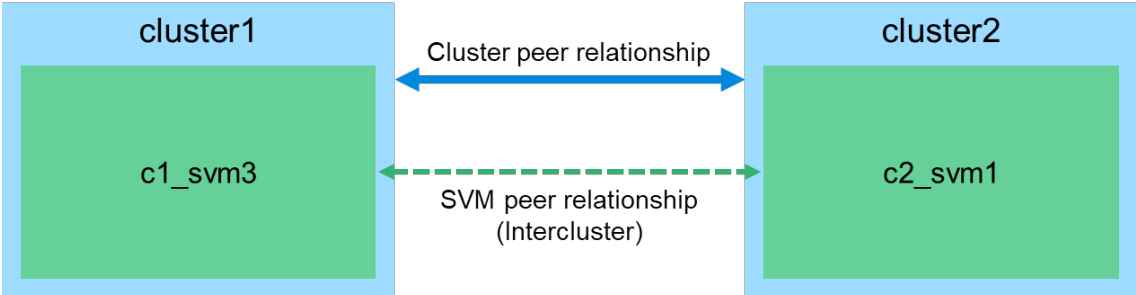
Step	Action
4-1	Perform the following steps for Cluster1: a. In the System Manager navigation pane, select Cluster > Settings. b. In the Intercluster settings > Storage VM peers pane, click the three vertical dots, and then select Peer storage VMs. Hosts Cluster Overview Settings Disks Support Intercluster settings Network interfaces IP addresses 192.168.0.106 192.168.0.105 Cluster peers Cluster2 Storage VM peers Add a storage VM peer to connect a storage VM on this cluster with one on a remote cluster. Add a storage VM peer
4-2	In the Peer storage VMs dialog box, provide the following information: Cluster1 Storage VM: c1_svm3 Cluster2 Storage VM: c2_svm1 Add a storage VM peer Local Cluster Cluster1 Storage VM c1_svm3 Remote Cluster Cluster2 Storage VM c2_svm1 Add a storage VM peer Cancel
4-3	Click Add a storage VM peer

Step	Action
4-4	Verify that the peering is successful and is listed in the Storage VM peers pane. 
4-5	For Cluster2, in the System Manager navigation pane, select Cluster > Settings.
4-6	In the Intercluster settings > Storage VM peers pane, verify that the peering is successful and is listed. 
4-7	 It can take a few seconds for the status to change.
4-8	In the Intercluster settings > Storage VM peers pane, click the three vertical dots, and then select Manage storage VM peers. 

Step	Action								
4-9	On the Storage VM peers page, verify that the storage VM peer relationship is listed and that the status is Peered. Storage VM peers Cluster settings + Peer storage VMs <table><tr><th>Storage VM</th><th>Peered cluster</th><th>Peered storage VM</th><th>Status</th></tr><tr><td>c2_svm1</td><td>Cluster1</td><td>c1_svm3</td><td>Peered</td></tr></table>	Storage VM	Peered cluster	Peered storage VM	Status	c2_svm1	Cluster1	c1_svm3	Peered
Storage VM	Peered cluster	Peered storage VM	Status						
c2_svm1	Cluster1	c1_svm3	Peered						
4-10	In Cluster 1, in the System Manager navigation pane, select Cluster > Settings.								
4-11	In the Intercluster Settings > Storage VM peers pane, click the three vertical dots, and then select Manage storage VM peers. Intercluster settings Network interfaces IP addresses ✓ 192.168.0.106 ✓ 192.168.0.105 Cluster peers ✓ Cluster2 Storage VM peers Peered storage ✓ 1 Add a storage VM peer Manage storage VM peers								
4-12	On the Storage VM peers page, verify that the storage VM peer relationship is listed and that the status is Peered. Storage VM peers Cluster settings + Peer storage VMs <table><tr><th>Storage VM</th><th>Peered cluster</th><th>Peered storage VM</th><th>Status</th></tr><tr><td>c1_svm3</td><td>Cluster2</td><td>c2_svm1</td><td>Peered</td></tr></table>	Storage VM	Peered cluster	Peered storage VM	Status	c1_svm3	Cluster2	c2_svm1	Peered
Storage VM	Peered cluster	Peered storage VM	Status						
c1_svm3	Cluster2	c2_svm1	Peered						

Task 5: Review the Exercise Environment

In this task, you review the exercise environment that you created in the previous tasks.

Step	Action
5-1	Verify that the following actions occurred: - You configured intercluster LIFs on each node of the two ONTAP clusters. - You established peer relationships between the Cluster1 and Cluster2 ONTAP clusters. - You created a peer relationship between storage VMs c1_svm3 and c2_svm1. 

End of exercise